Python(22AIE205) Programming Lab

Weekly Lab Report-2

LAB 1: IF ELSE ,IF ELSE IF,Nested IF Control statements.

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Experiment Date:	

Experiment Record:

Date:	21-07-2026
Experiment Name:	IF ELSE ,IF ELSE IF,Nested IF Control statements
Aim:	To practice IF ELSE ,IF ELSE IF,Nested IF Control statements using Python IDLE.
Software Required:	Python 3.x and Python IDLE

Questions Given:

1.Write a Python program that accepts a student's marks (out of 100) and determines the grade based on the slab table below. The program should also validate the input and display the pass/fail status.

2.An e-commerce company wants to automate discount calculation for its customers during a festive sale. Write a Python program that accepts the

total purchase amount and the customer's membership type, then calculates the final payable amount according to the rules below..

3. A power distribution company has introduced a new billing policy to encourage responsible electricity usage. Write a Python program that accepts the number of electricity units consumed in a month and calculates the total electricity bill according to the slab rules below.

Programs, Sample Outputs and Results:

Question 1: Student Grade Calculator

Problem Statement: Write a Python program that accepts a student's marks (out of 100) and determines the grade based on the slab table below. The program should also validate the input and display the pass/fail status.

Program:

grade1.py - C:\Users\prana\OneDrive\Desktop\Lab2\grade1.py (3.14.6)

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```
while True:
 a=int(input("Enter Your Marks:"))
 if (a<100)and(a>=90):
     print("The grade is A+ and marks ontained:",a)
     print("Remark:Outstanding")
     print("Result:Pass")
 elif (a<90)and(a>=75):
     print("The grade is A and marks ontained:",a)
     print("Remark:Excellent")
     print("Result:Pass")
 elif (a<75)and(a>=60):
     print("The grade is B  and marks ontained:",a)
     print("Remark: Good")
     print("Result:Pass")
 elif (a<60)and(a>=50):
     print("The grade is C and marks ontained:",a)
     print("Remark:Average")
     print("Result:Pass")
 elif (a<50)and(a>=40):
     print("The grade is D and marks ontained:",a)
     print("Remark:Pass")
     print("Result:Pass")
 else:
     print("grade is F")
     print("Remark:Fail")
```

Input and Output:

```
Enter "help" below or click "Help" above for more information.
>>>
============ RESTART: C:\Users\prana\OneDrive\Desktop\Lab2\grade1.py ====
Enter Your Marks:89
The grade is A and marks ontained: 89
Remark:Excellent
Result:Pass
Enter Your Marks:
```

Result: The program accepted and displayed the student Grade by their Marks successfully using if elif control statements.

Question-2: Online Shopping Discount Calculator.

Problem Statement: An e-commerce company wants to automate discount calculation for its customers during a festive sale. Write a Python program that accepts the total purchase amount and the customer's membership type, then calculates the final payable amount according to the rules below..

Program:

```
while True:
  a = int(input("Enter the Purchased Amount: "))
  b = int(input("Enter Membership type 1.Regular 2.Silver 3.Gold 4.Platinum: "))

  if b == 1:
    if a < 500:
      print("No discount")
    else:
      print("Original Amount:", a)
      print("Discount Amount:", (a * 5) / 100)
      d = (a * 5) / 100

      if (a - d) < 1000:
        print("Amount Should Pay:", a - d + 50)
      else:
        print("Amount Should Pay:", a - d)

  elif b == 2:
```

```
    if a > 10000:
        print("Original Amount:", a)
        print("Discount Amount:", (a * 10) / 100 + 500)
        e = (a * 10) / 100 + 500

        if (a - e) < 1000:
            print("Amount Should Pay:", a - e + 50)
        else:
            print("Amount Should Pay:", a - e)

    elif (a < 10000) and (a >= 500):
        print("Original Amount:", a)
        print("Discount Amount:", (a * 10) / 100)
        e = (a * 10) / 100

        if (a - e) < 1000:
            print("Amount Should Pay:", a - e + 50)
        else:
            print("Amount Should Pay:", a - e)

    else:
        print("No discount")

elif b == 3:
    if a > 10000:
        print("Original Amount:", a)
        print("Discount Amount:", (a * 15) / 100 + 500)
        f = (a * 15) / 100 + 500

        if (a - f) < 1000:
            print("Amount Should Pay:", a - f + 50)
        else:
            print("Amount Should Pay:", a - f)
```

```
    elif (a < 10000) and (a >= 500):
        print("Original Amount:", a)
        print("Discount Amount:", (a * 15) / 100)
        f = (a * 15) / 100

        if (a - f) < 1000:
            print("Amount Should Pay:", a - f + 50)
        else:
            print("Amount Should Pay:", a - f)

    else:
        print("No discount")

elif b == 4:
    if a > 10000:
        print("Original Amount:", a)
        print("Discount Amount:", (a * 20) / 100 + 500)
        g = (a * 20) / 100 + 500

        if (a - g) < 1000:
            print("Amount Should Pay:", a - g + 50)
        else:
            print("Amount Should Pay:", a - g)

    elif (a < 10000) and (a >= 500):
        print("Original Amount:", a)
        print("Discount Amount:", (a * 20) / 100)
        g = (a * 20) / 100

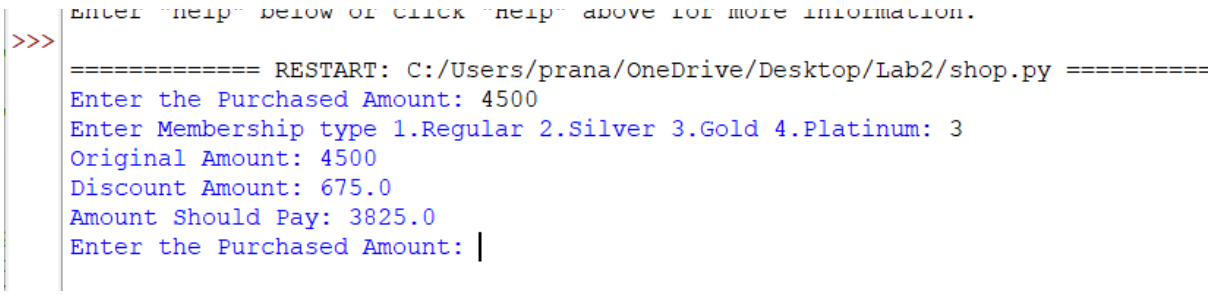
        if (a - g) < 1000:
            print("Amount Should Pay:", a - g + 50)
        else:
```

```
        print("Amount Should Pay:", a - g)


    else:
        print("No discount")


else:
    print("Invalid Membership Type")
```

Input&Output:



Result:The program is executed successfully ,by taking input from user and applied all conditions.

Question-3:Electricity Bill Generator (Slab System)

Problem Statement: A power distribution company has introduced a new billing policy to encourage responsible electricity usage. Write a Python program that accepts the number of electricity units consumed in a month and calculates the total electricity bill according to the slab rules below.

Program:

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```
units = int(input("Enter units consumed: "))

bill = 0

if units <= 100:
    bill = units * 2
elif units <= 200:
    bill = (100 * 2) + (units - 100) * 3.5
elif units <= 300:
    bill = (100 * 2) + (100 * 3.5) + (units - 200) * 4.5
else:
    bill = (100 * 2) + (100 * 3.5) + (100 * 4.5) + (units - 300) * 6

if bill < 200:
    bill = 200

if units <= 300:
    discount = 0
elif units <= 500:
    discount = bill * 0.05
else:
    discount = bill * 0.10

final_bill = bill - discount

print("Units Consumed :", units)
print("Bill Before Discount : Rs", bill)
print("Discount Amount : Rs", discount)
print("Final Payable Amount : Rs", final_bill)
```

Input & Output:

```
============= RESTART: C:/Users/prana/OneDrive/Desktop/Lab2/bill.py =======
Enter units consumed: 50
Units Consumed : 50
Bill Before Discount : Rs 200
Discount Amount : Rs 0
Final Payable Amount : Rs 200
```

4. Observations

- The **if, elif, and else** statements are used to make decisions based on conditions.
- **Nested if** statements allow one condition to be checked inside another, which is useful when a decision depends on more than one factor (such as membership type and amount).
- Comparison operators such as **>, <, >=, <=, ==, and !=** are used to compare values in conditional statements.
- **Indentation** is mandatory in Python to define the block of code under **if, elif, and else**.

- The **input()** function accepts user input as a string, so **int()** is used for numerical comparisons.
- Python IDLE displays the program output immediately after execution, making it easy to verify different test cases.

5. Overall Result

Both the assigned programs based on **if-elif-else** and **nested if** statements were written, executed, and verified successfully using Python IDLE. The outputs matched the expected results for the given sample inputs, demonstrating the correct implementation of decision-making statements in Python.